

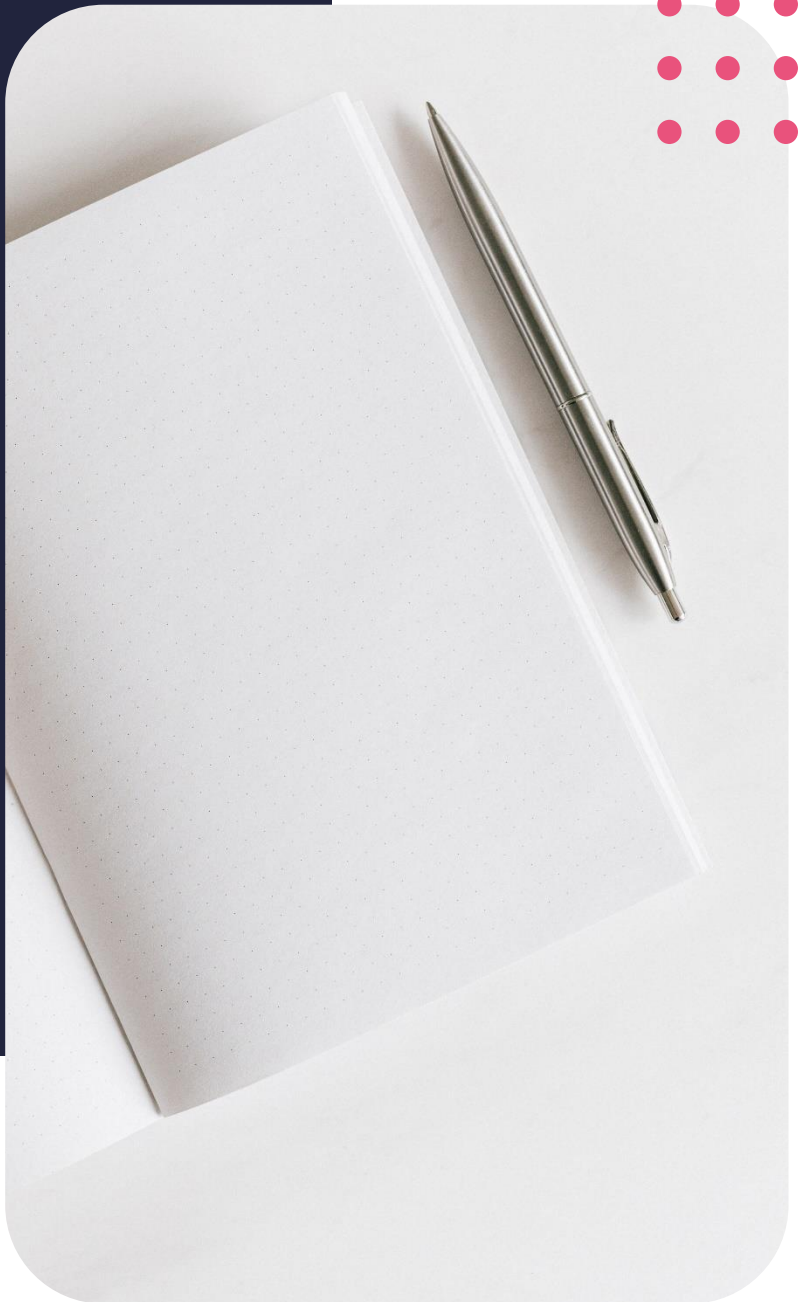
Diploma in Data Analytics

Time series analysis



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Lesson outcomes

By the end of this lesson, you should be able to:

- Understand, recognise, and decompose the concepts of time series analysis
- Perform some time series analysis functions in R

Introduction

Time series analysis considers data points taken over a period that may have autocorrelation, trends or seasonal variation that needs to be accounted for. Time series and the analysis therefore is becoming increasingly more important to our generation due to the large creation of data through various sources like the internet of things, digitization of healthcare and the evolution of smart cities.

This lesson will delve deeper into time series analysis. We will further discuss the concepts surrounding time series analysis that we introduced in the previous lesson and add some new concepts to that. Thereafter we will break down and understand some of the time series models a bit better. We will end today's lesson by looking at how we can apply these concepts learnt in R in a more practical sense.

Time series concepts

Time series recap

In the previous lesson we learnt that time series is:

A variable that is measured over time in sequential order and is called a time series. It may consist of weekly production inputs, monthly sales, annual rainfall, or any other variable for which the value is observed at regular intervals.

Through the analysis of data that spans over a period of time, we are able to spot trends and patterns and in these will aid us to recognize variations from the past in order to more accurately predict future instances.

We also introduced some time series terms in the last lesson:

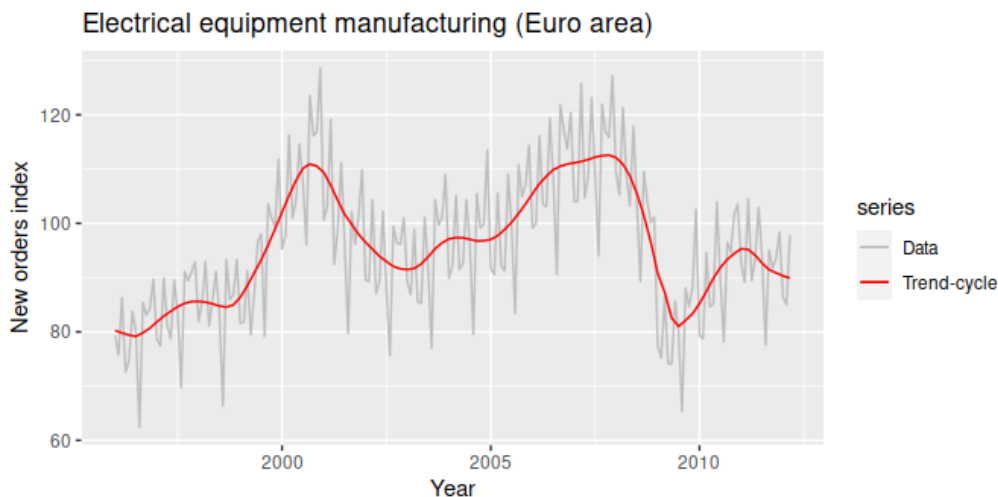
- Autocorrelation is also known as the degree of similarity between observations as a function of the time lag amongst them.
- Seasonality refers to the existence of variations that take place at specific methodical intervals less than a year apart, for instance weekly, monthly, or quarterly.
- Time series is assumed to be stationary if the mean, variance, and covariance does not change over time.

Components of time series

More components that exist as a part of time series analysis, is the four possible ways to combine a time series model. These components can be combined either multiplicatively, additively or in combination of both these approaches.

1. Trend variations

- Trend variations show us the general tendency of the data to increase or decrease over a long-term period. It can be defined as a long term smooth underlying movement in time.
- Long term meaning, that the duration is more than a year.
- Examples include population growth, urbanisation, and technological improvements.



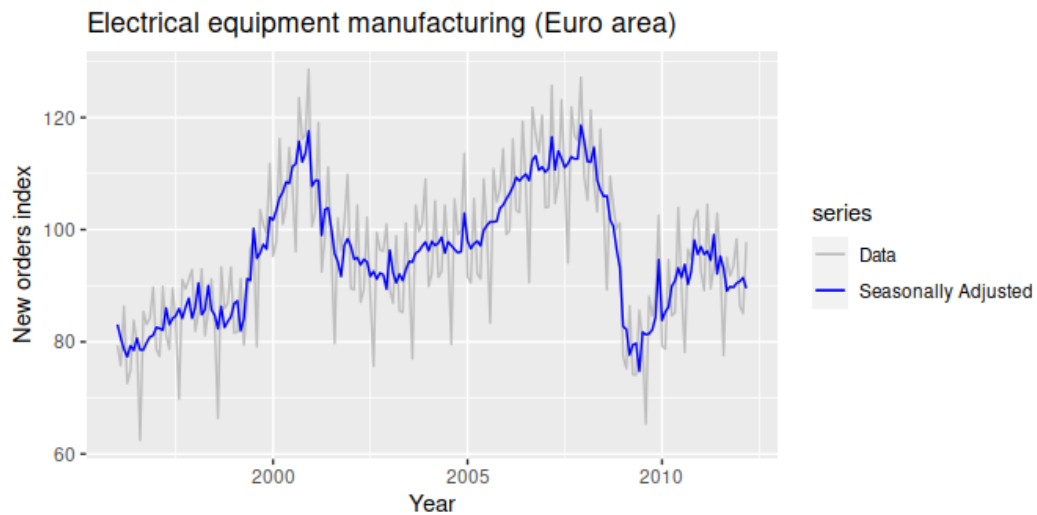
- The figure is an illustration of an upward trend variation over a long-term period. The data represents the number of new orders for electrical equipment in 16 countries. The trend cycle shows us the overall movement of the data ignoring small random and seasonal fluctuations.

2. Cyclical variations

- Cyclical variations are variations that repeat over time.
- The period it repeats over is more than a year from one peak to the next.
- Examples could include actions of government bodies that lead to the public being more optimistic or pessimistic about the future of the country and in turn leads to an increase or decrease in economic levels.
- The cause of this type of cycle is therefore difficult to predict.

3. Seasonal variation

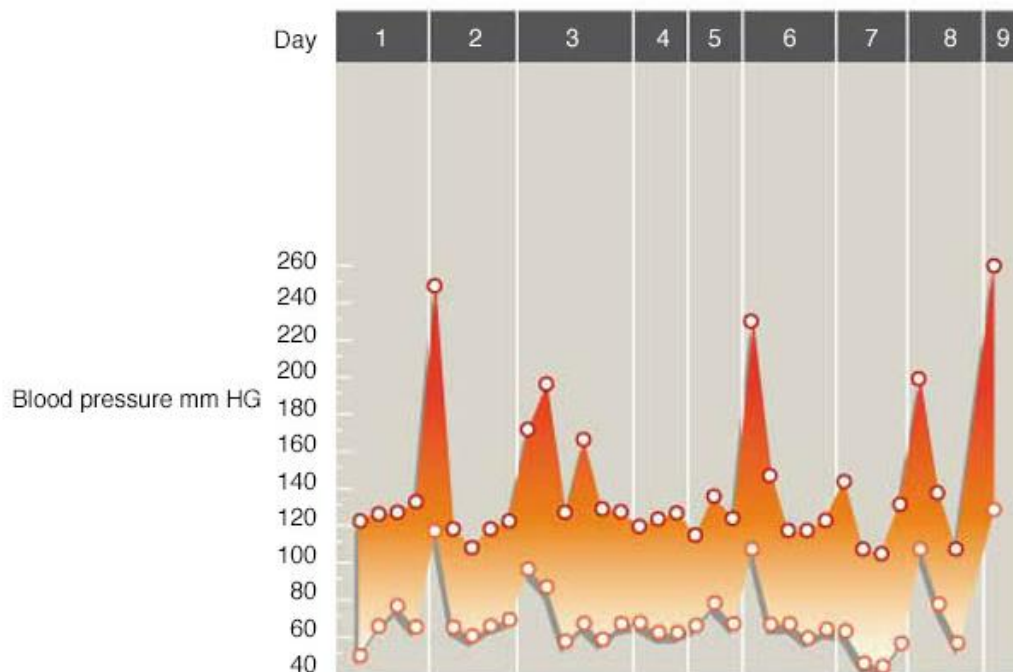
- We have already discussed seasonal variations in the previous lesson, but to recap, this type of variation are fluctuations that repeat over a period of less than a year.
- Examples include components that are used to represent a regularly repeating pattern that is less than one year in duration.
- This could be daily, weekly, monthly, or quarterly.
- These variations are caused by climate changes, religious events, public and school holidays



- e. This figure shows that we are able isolate the seasonal variation components of the data by adjusting for it as seen here on this plot, but more on this a bit later in the lesson.

4. Irregular variations

- Lastly, irregular variations can occur from random or chance events and is represented by random fluctuations in the data.
- These types of fluctuations are unpredictable and are caused by events like floods, disasters, strikes and so forth.
- They follow no pattern and can make it more difficult for us to identify the effect of other components.



- d. The figure above is an example of a random variation caused by a pheochromocytoma which is a rare non-cancerous tumour that grows on the adrenal glands. The tumour releases hormones that can cause random fluctuations in blood pressure that can make the condition very difficult to detect. This plot shows a nine-day period of short, random bursts in blood pressure due to pheochromocytoma.

Time series models

The next topic will touch on understanding how to decompose time series to analyse the trend component and investigate seasonal analysis in more depth.

Decomposing

Two methods that can be used in time series analysis to evaluate the trend component:

- Regression techniques to fit a trend line to the data
- Moving average method

Regression techniques

Time series regression techniques are often used in the analysis of financial data. Regression techniques can be used to assess the relationship between financial variables. Time series regression techniques can also be used to substantiate the informational effectiveness of the financial markets.

The trend line for regression techniques can be either linear or quadratic.

Moving averages method

The motivation behind the moving average technique is to remove the short term seasonal and abnormal variations, to expose the combined trend and cyclical components.

The moving average method smoothed out any short-term fluctuations the data might contain.

The moving average for a time is the mean of the values in that period and those values close to it. Mathematically, the moving average can be represented by the sum of the most recent k data values divided by k , where k is the period number.

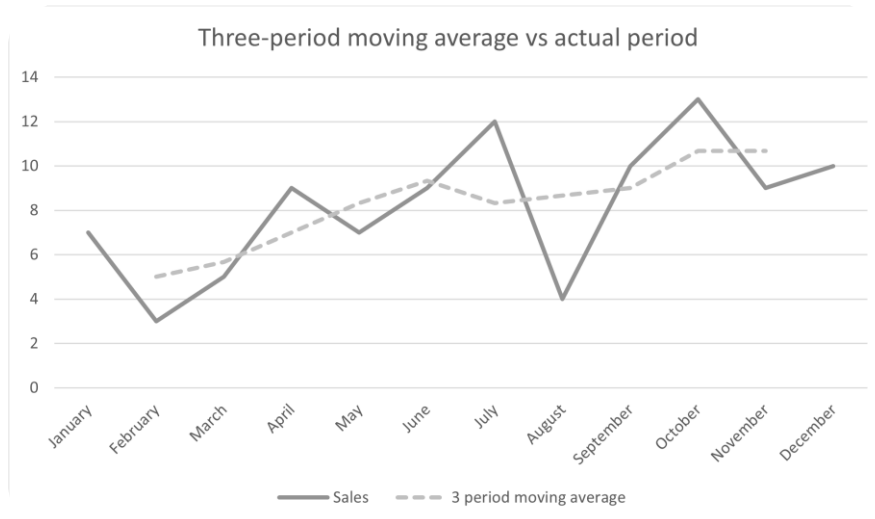
2010	Sales	2010	Sales
January	7	July	12
February	3	August	4
March	5	September	10
April	9	October	13
May	7	November	9
June	9	December	10

This table represents the monthly sales of the company from January to December for the year 2010.

To calculate the moving average for the first three-month period, we would group the monthly sales of the first three periods, January, February, and March together and then average them. The average for these periods would therefore be 5. The moving average of 5 corresponds to February, the middle period.

We calculate the second moving average to March for monthly sales by calculating the average of February, March, and April. The average for these three periods is 5,67 approximately. The second moving average is calculated by dropping the first period and adding the fourth.

In the same way we calculated the average to February and March, the same way we will calculate the average for March, April and May and achieve an average of 7 corresponding to April. The process will continue until we eventually calculate three-period moving averages for February through to November. January and December do not have three-period moving averages because January does not have any preceding sales and December does not have any succeeding sales.



When we plot these three periods moving averages against the actual data, we can see that the moving average presents a much smoother, less fluctuating trend, than that of the original data. If we calculated the moving average over a five-period, the curve would have been smoother than that of the three-period moving average. A moving average can therefore help us to smooth out the curve for short-term fluctuations to better evaluate long term trends.

Seasonal analysis

Seasonal analysis is present when seasonal patterns arise over specific periods in the year. We apply a seasonal index factor to this pattern that will inflate or deflate a trend value to compensate for the effect of the pattern that repeats itself over the period of a year or less.

Deseasonalising

In the process of applying the seasonal index, we remove the seasonal influence from the time series. Through deseasonalisation, we can more easily account for the influence of the trend variations, cyclical variations, and irregular variations in the data.

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